

4.4 L'Hôpital's rule and the inverse function theorem.

1. Evaluate the limit.

(a) $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^4}$

(c) $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x}$

Miscellaneous

2. Prove that

$$e^{\sin(x)} \leq 1 + ex,$$

for all $x \geq 0$.

3. Prove that if $f : (a, b) \rightarrow \mathbb{R}$ satisfies

$$|f(y) - f(x)| \leq (y - x)^2, \quad \forall x, y \in (a, b),$$

then f is constant.

Although Integrals are not part of the syllabus, it doesn't hurt to use what you (allegedly) already know. Feel free to use the FTC on the next question.

Theorem 1 (Fundamental Theorem of Calculus). Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Let $F : [a, b] \rightarrow \mathbb{R}$ be defined by

$$F(x) = \int_a^x f(t) dt.$$

Then F is uniformly continuous on $[a, b]$ and differentiable on (a, b) , and

$$F'(x) = f(x)$$

for all $x \in (a, b)$.

4. Show that if

$$\frac{a_0}{1} + \frac{a_1}{2} + \dots + \frac{a_n}{n+1} = 0,$$

then the equation

$$a_0 + a_1x + \dots + a_nx^n = 0$$

has a zero in the interval $(0, 1)$.